ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART B)

Topic 6b: AC Circuit Analysis

AC Power Analysis

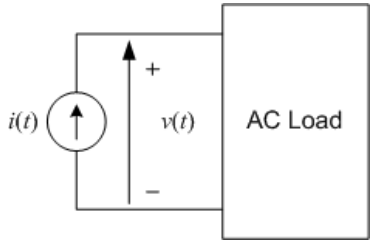
- ✓ AC Power
- ✓ Reactive Power
- ✓ Apparent Power
- ✓ Power Triangle
- ✓ Maximum Power Transfer
- ✓ Power Factor Correction

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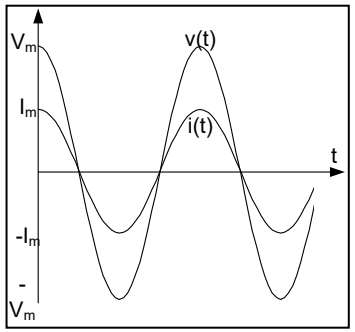
AC Power - Current, Voltage and Power Transfer for a Resistive Load

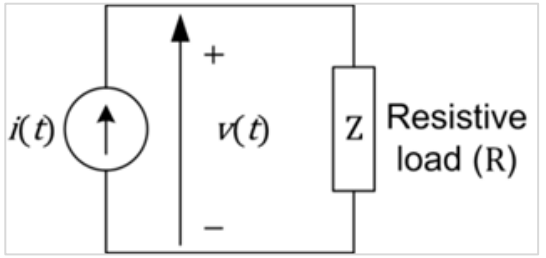
- Applying an AC signal to a circuit containing resistive as well as reactive elements results in power transfer from the source to the circuit.



- Consider the case that the load is a pure resistance

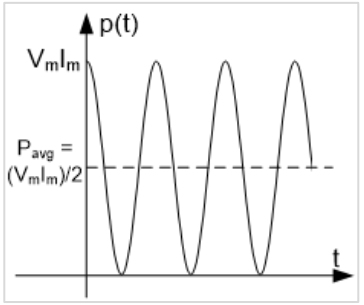
$$v(t) = V_m \cos(\omega t)$$
$$i(t) = I_m \cos(\omega t)$$





- The instantaneous power

$$p(t) = v(t) \cdot i(t)$$
$$= V_m I_m \cos^2(\omega t)$$ ← always positive



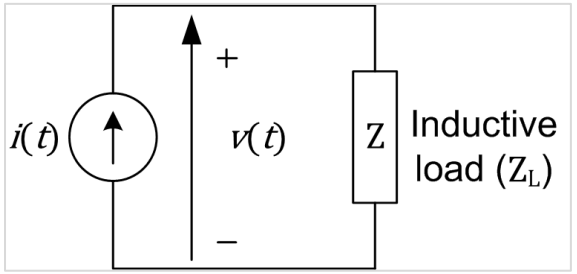
- Because $p(t)$ is positive at all times, the energy flows continually in the direction from the source to the load (where it is converted to heat).

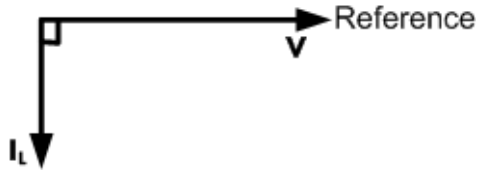
AC Power - Current, Voltage and Power Transfer for an Inductive Load

- Consider the case where the load is a pure inductance for which $z = j\omega L = \omega L \angle 90^o$

$$v(t) = V_m \cos(\omega t)$$

$$i(t) = \frac{v(t)}{Z} = I_m \cos(\omega t - 90^o) = I_m \sin(\omega t)$$

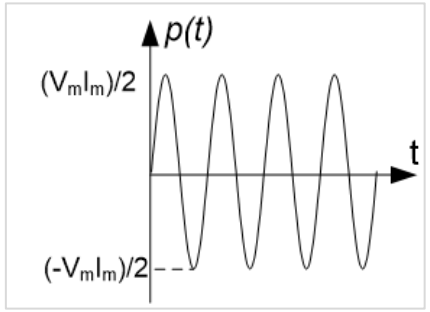




The current through the inductor lags the voltage across the inductor by 90°

$$p(t) = v(t) \cdot i(t) = V_m I_m \cos(\omega t) \sin(\omega t)$$

$$= \frac{V_m I_m}{2} \sin(2\omega t)$$ ← average power = 0



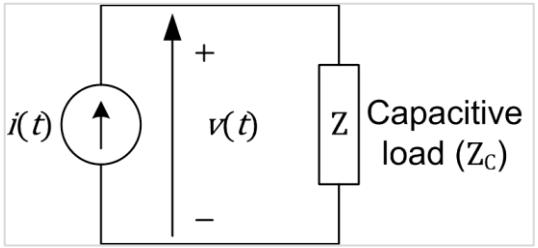
- Half the time the power is positive, showing that the energy is delivered to the inductance, where it is stored in the magnetic field.
- For the other half of the time, power is negative, showing that the inductance return energy to the source.
- Notice that the average power is zero. In this case, the reactive power flows from the source to the load

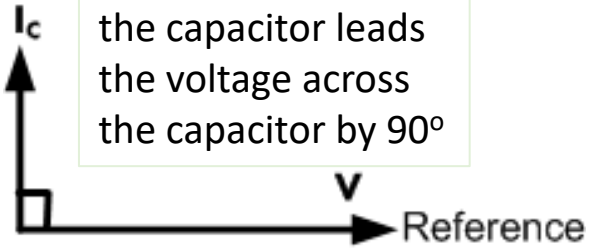
AC Power - Current, Voltage and Power Transfer for a Capacitive Load

- Consider the case where the load is a pure capacitance for which $Z=1/j\omega C=\{1/\omega C\}\angle -90^{o}$

$$v(t) = V_m \cos(\omega t)$$

$$i(t) = \frac{v(t)}{Z} = I_m \cos(\omega t + 90^{o}) = -I_m \sin(\omega t)$$

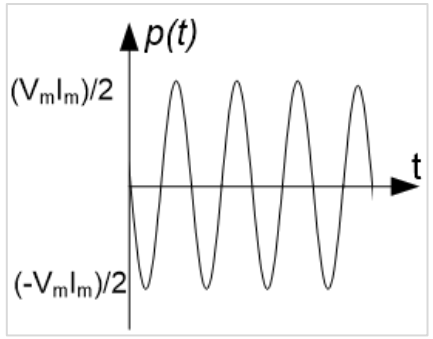




The current through the capacitor leads the voltage across the capacitor by 90°

$$p(t) = v(t) \cdot i(t) = V_m I_m \cos(\omega t) \sin(\omega t)$$

$$= \frac{V_m I_m}{2} \sin(2\omega t)$$ ← average power = 0

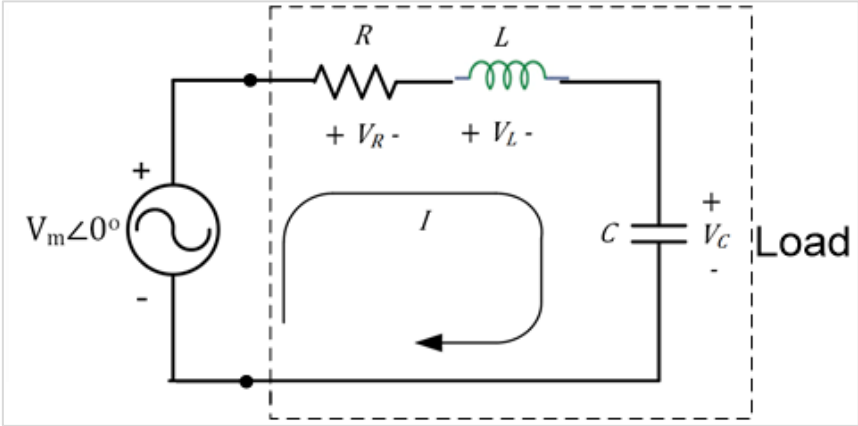


- The power for the capacitance carries the opposite sign as that for the inductance. We say that the reactive power flows from the load to the source.

- Thus, we say that reactive power is positive for an inductance and negative for a capacitance.

AC Power – Load Contains both Inductance and Capacitance

- If the load contains both inductance and capacitance with reactive powers of equal magnitude, the reactive powers cancel (see example below)



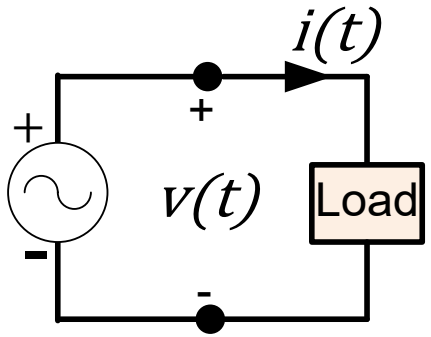
- The power flows back and forth to inductances and capacitances is called reactive power.
- Even though **no** average power is consumed by a pure energy-storage element (inductance or capacitance), reactive power is still important.
- It is possible to have loads composed of energy-storage elements that draw large currents requiring heavy-duty wiring, even though little average power is consumed.
- Therefore, electric power delivery companies charge their customers for reactive power (but at a lower rate) as well as for total energy delivered.

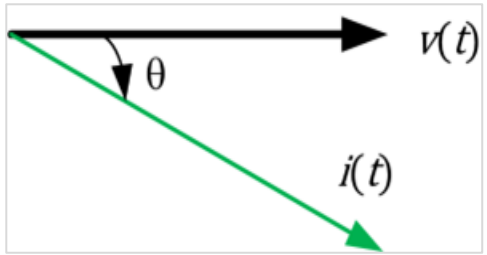
Power Calculation for a General Load

- Now, let us consider the voltage, current, and power for a general RLC load, for which the phase θ can be any value from -90° to +90°

$$v(t) = V_m \cos(\omega t)$$

$$i(t) = I_m \cos(\omega t - \theta)$$





For a purely resistive load θ = 0.

- The instantaneous power $p(t)$

$$p(t) = v(t) \cdot i(t) = V_m I_m \cos(\omega t)\cos(\omega t - \theta)$$

$$= V_m I_m \cos(\omega t)\big(\cos(\omega t)\cos(\theta) + \sin(\omega t)\sin(\theta)\big)$$

$$= V_m I_m \cos(\theta)\cos^2(\omega t) + V_m I_m \sin(\theta)\sin(\omega t)\cos(\omega t)$$

$$= V_m I_m \cos(\theta)\left[\frac{1+\cos(2\omega t)}{2}\right] + \frac{V_m I_m}{2}\sin(\theta)\sin(2\omega t)$$

$$= \frac{V_m I_m}{2}\cos(\theta)[1 + \cos(2\omega t)] + \frac{V_m I_m}{2}\sin(\theta)\sin(2\omega t)$$

Power Calculation for a General Load

- Notice that, the term involving $\cos(2\omega t)$ and $\sin(2\omega t)$ have average values of zero. Thus the **average power (P)** is

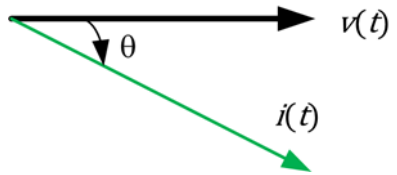
$$P = \frac{V_m I_m}{2}\cos(\theta) = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos(\theta)$$

$$p(t) = \frac{V_m I_m}{2}\cos(\theta)[1+\cos(2\omega t)] + \frac{V_m I_m}{2}\sin(\theta)\sin(2\omega t)$$

$$P = V_{rms} I_{rms}\cos(\theta) \quad \text{Watts}$$

- The term $\cos(\theta)$ is called the power factor (*pf*).

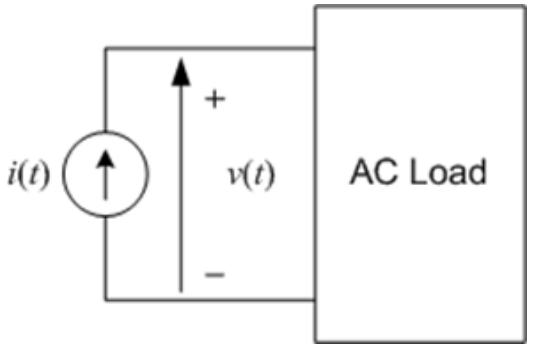
$$pf = \cos(\theta)$$



- Sometimes θ is called the power angle.
- It is common to state that the current leads (in capacitive loads; i.e leading power factor) or lags (in inductive loads; i.e. lagging power factor) the voltage.
- A typical power factor would be stated to be 90% lagging which means that $\cos(\theta)=0.9$ and the current lags the voltage.

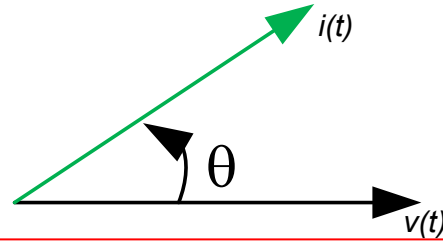
Power factor

- Assuming voltage $v(t)$ as the reference, look at the current $i(t)$ relative to voltage.



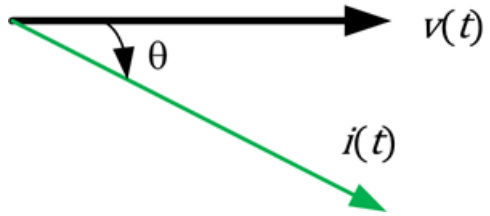
1

✓ If the current is leading the voltage, i.e leading power factor which implies a capacitive load



2

✓ If the current is lagging the voltage, i.e lagging power factor which implies an inductive load.



3

Note: We assume a voltage having zero phase. In general, the phase of the voltage may have a value other than zero. Then θ should be taken as the phase of the voltage θ_v minus the phase of the current θ_i

$$\theta = \theta_v - \theta_i$$

Power factor (*PF*) is the cosine of the angle θ by which the current lags the voltage.

Reactive Power

- The power flowing back and forth between the source and the energy-storage element is the last term
  The instantaneous power $p(t)$:

$$p(t) = \frac{V_m I_m}{2}\cos(\theta)[1 + \cos(2\omega t)] + \frac{V_m I_m}{2}\sin(\theta)\sin(2\omega t)$$

In ac circuits energy flows into and out of energy storage elements (inductances and capacitances)

- The peak value is defined to be the **reactive power** Q.

$$Q = \frac{V_m I_m}{2}\sin(\theta)$$

$$Q = V_{rms} I_{rms}\sin(\theta) \quad \text{VAR}$$

- For a purely resistive load $\theta = 0$; $\therefore Q = 0$

- Q does not represent the flow of net energy and its units are usually given as VARs (Volt Ampere Reactive)

Apparent Power

- The apparent power which is defined as the product of the effective voltage (V_{rms}) and effective current (I_{rms}).

Apparant power = $V_{rms}\, I_{rms}$

- Its unit is Volt-Ampere (VA).

- We have P *(average power)* and Q *(reactive power)*

$$P = V_{rms} I_{rms} \cos(\theta)$$

$$Q = V_{rms} I_{rms} \sin(\theta)$$

$$P^2 + Q^2 = (V_{rms} I_{rms})^2 \left(\cos^2(\theta) + \sin^2(\theta)\right)$$

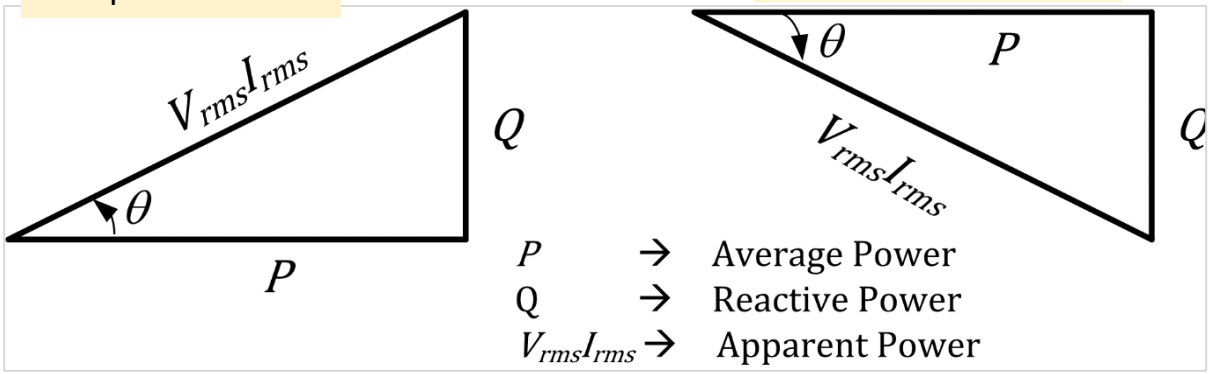
$$P^2 + Q^2 = (V_{rms} I_{rms})^2$$

- If we say that we have a 5 kW load, this means that P = 5 kW.
- If we have 5kVA load, $V_{rms}I_{rms}$ = 5kVA
- If we say that a load absorbs 5 kVAR then Q = 5 kVAR

Power Triangle

- The relationship between real power or average power (P), reactive power (Q), apparent power $V_{rms}I_{rms}$ and the power angle can be represented by the **power triangle**.
- For an **inductive load** , in which case θ [$\theta = \theta_v$- θ_i] and the reactive power Q are positive and the power triangle is shown below.
- The power triangle for a **capacitive load is** shown in which case θ [$\theta = \theta_v$- θ_i] and the reactive power Q are negative.

For inductive load, θ is positive



For capacitive load, θ is negative

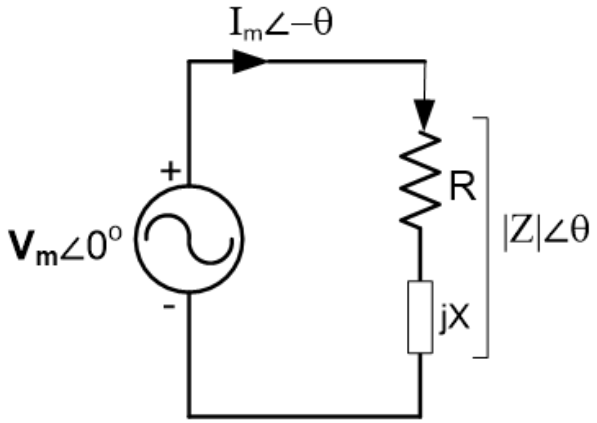
P → Average Power
Q → Reactive Power
$V_{rms}I_{rms}$ → Apparent Power

$$P = V_{rms}I_{rms}\cos(\theta)$$

$$Q = V_{rms}I_{rms}\sin(\theta)$$

Power in AC circuits: additional relationships

- Consider the circuit shown below where a voltage source delivering power to a load impedance z.



$$\boldsymbol{I} = \frac{\boldsymbol{V}}{z} = \frac{V_m\angle 0^o}{|z|\angle\theta} = I_m\angle -\theta \quad where\ z = R + jX$$

$$\cos(\theta) = \frac{R}{|z|} \quad \text{and} \quad \sin(\theta) = \frac{X}{|z|} \quad \text{and} \quad I_m = \frac{V_m}{|z|}$$

- We have $P = \frac{V_m I_m}{2}\cos(\theta) = \frac{V_m I_m}{2}\frac{R}{|z|} = \frac{{I_m}^2 R}{2} = {I_{rms}}^2 R$

- In a similar fashion we can show that $Q = {I_{rms}}^2 X$

- For inductive load X is positive, where as for a capacitive load , X is negative.

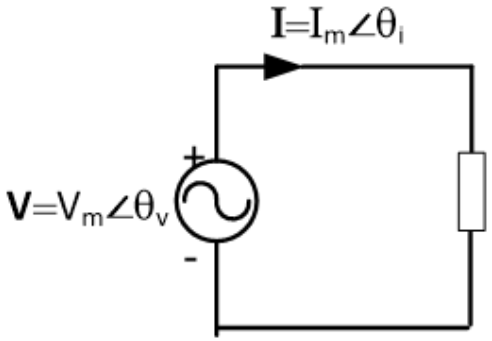
- Furthermore , we can show that $P = \frac{{V_{Rrms}}^2}{R} \quad and \quad Q = \frac{{V_{Xrms}}^2}{X}$

V_{Rrms} is the rms value of the voltage across the resistance
and V_{Xrms} is the rms value of the voltage across the reactance

Complex Power

- Consider the circuit shown below. The complex power, denoted by ***S,*** delivered to the circuit is given by

$$\boldsymbol{S} = \frac{1}{2}\boldsymbol{VI}^*$$



$$\boldsymbol{S} = \frac{1}{2}\boldsymbol{VI}^* = \frac{1}{2}\,(\mathrm{V_m}\angle\theta_v)(\mathrm{I_m}\angle-\theta_i) = \frac{V_m I_m}{2}\angle\theta_v - \theta_i = \frac{V_m I_m}{2}\angle\theta$$

- where as before $\theta = \theta_v - \theta_i$ is the power angle.

$$\boldsymbol{S} = \frac{1}{2}\boldsymbol{VI}^* = \frac{V_m I_m}{2}\cos(\theta) + j\frac{V_m I_m}{2}\sin(\theta) = P + jQ$$

- The first term on the right-hand side is the average power P delivered to the circuit and the second term is j times the reactive power Q

- If we know the complex power S, then we can find the average power, reactive power and apparent power:

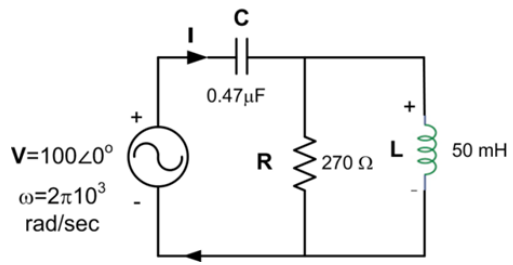
$$\boldsymbol{P} = \mathrm{Re}(\boldsymbol{S}) = \mathrm{Re}\left(\frac{1}{2}\mathbf{VI}^*\right);\ \boldsymbol{Q} = \mathrm{Im}(\boldsymbol{S}) = \mathrm{Im}\left(\frac{1}{2}\mathbf{VI}^*\right);$$

$$\text{Apparent power} = |\boldsymbol{S}| = |\frac{1}{2}\mathbf{VI}^*|$$

Examples

Example 1

❑ Find the apparent power as well as the average power delivered by the source as shown on the right. What is the power factor of this circuit? Is it leading or lagging?



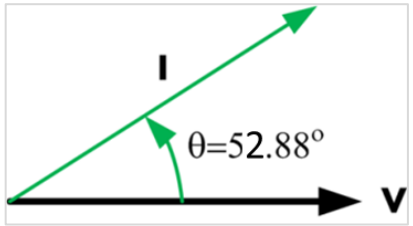
Solution

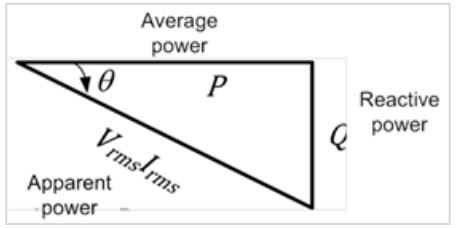
$$Z = Z_C + Z_R||Z_L = \frac{1}{j\omega C} + \frac{j\omega LR}{R + j\omega L}$$

$$= \frac{-j}{2\pi 10^3 \times 0.47 \times 10^{-6}} + \frac{j2\pi 10^3 \times 0.05 \times 270}{270 + j2\pi 10^3 \times 0.05}$$

$$= 155.3j - j205.2 = 257.3\angle -52.88^o$$

$$I = \frac{V}{Z} = \frac{100\angle 0^o}{257.3\angle -52.88^o} = 0.3886\angle 52.88^o$$





- Total current is leading to total voltage. Hence, Capacitive load; therefore leading power factor

$$Power\ factor(pf) = \cos(\theta) = \cos(52.88^o) = 0.6035 \quad (leading)$$

$$Apparent\ Power = V_{rms}I_{rms} = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}} = \frac{100 \times 0.3886}{2} = 19.43\ \text{VA}$$

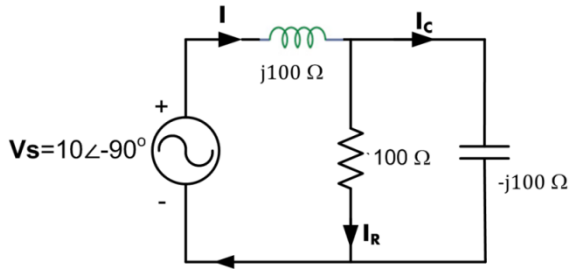
$$P_{average} = V_{rms}I_{rms}\cos(\theta) = 19.43 \times 0.6035 = 11.72\ \text{Watts}$$

- The source delivers an apparent power of 19.43 VA, however only 11.72 W (or 60.35%) of the apparent power is dissipated as heat

Examples

Example 2

❑ Compute the power and reactive power taken from the source for the circuit shown below. Also calculate the power and reactive power delivered to each element in the circuit.



Solution

- The currents were computed in Topic 6: $\boldsymbol{I} = 0.1414\angle -135^{o}$; $\boldsymbol{I_R} = 0.1\angle -180^{o}$; $\boldsymbol{I_C} = 0.1\angle -90^{o}$.
- To find the power and reactive power, we must find the power angle, θ, which is the angle between the input source voltage (V_s)and the current delivered (I).

$$\theta = \angle -90^{o} - \angle -135^{o} = 45^{o};\ V_{rms} = \frac{|V_S|}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 7.071;\ I_{rms} = \frac{|I|}{\sqrt{2}} = \frac{0.1414}{\sqrt{2}} = 0.1$$

$$P_{average} = V_{rms} I_{rms} \cos(\theta) = 7.071 \times 0.1 \times \cos(45^{o}) = 0.5\ Watts$$

$$Reactive\ Power\ (Q) = V_{rms}\, I_{rms} \sin(\theta) = 7.071 \times 0.1 \times \sin(45^{o}) = 0.5\ VAR$$

- Reactive power delivered to the inductor and capactior:

$$Q_L = I_{rms}^2 X_L = (0.1)^2 \times 100 = 1.0\ VAR$$

$$Q_C = I_{c}^2 rms X_C = \left(\frac{0.1}{\sqrt{2}}\right)^2 \times (-100) = -0.5\ VAR$$

- Reactive power delivered by the source is equal to the sum of the reactive power absorbed by the inductor and by the capacitor:

$$Q = Q_L + Q_C = 1.0\ VAR - 0.5\ VAR = 0.5 VAR$$

$$P_{average} = I_{R}^2 rms R = \left(\frac{0.1}{\sqrt{2}}\right)^2 \times 100 = 0.5\ Watts$$

- Of course, the power absorbed by the capacitance and inductance is zero, $P_L = 0$, $P_C = 0$. All of the power delivered by the source is absorbed by the resistance.

Examples

Solution contd…

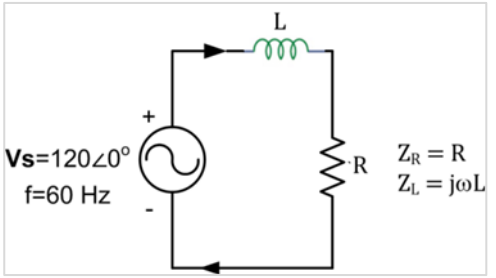
- An alternative method for computing *P* and *Q* is to first find the complex power and then take real and imaginary parts:

$$\boldsymbol{S} = \frac{1}{2}\boldsymbol{V}\boldsymbol{I}^* = \frac{1}{2}(10\angle -90^o)(0.1414\angle 135^o) = 0.707\angle 45^o = 0.5 + j0.5$$

$$\mathrm{P} = \mathrm{Re}(\mathbf{S}) = 0.5\ \mathrm{W};\quad \mathrm{Q} = \mathrm{Im}(\mathbf{S}) = 0.5\ \mathrm{VAR}$$

Example 3

❑ A series of RL circuit is connected to the household AC line. If the voltage across the inductance is 60V (rms) and the average power dissipated by the circuit is 25 W, what are the values of R, L, and power factor (pf).

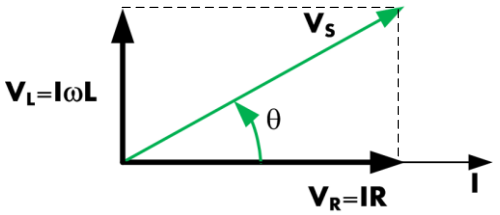


Solution

$$V_L = \left|\frac{Z_L}{Z_R + Z_L}\right| V_S;\quad \frac{60}{120} = \left|\frac{j\omega L}{R + j\omega L}\right|;\quad 0.5 = \frac{\omega L}{\sqrt{R^2 + \omega^2 L^2}};\quad 0.25 = \frac{\omega^2 L^2}{R^2 + \omega^2 L^2};\ \therefore\ \omega L = \frac{R}{\sqrt{3}}$$

$$\tan\theta = \frac{\omega L}{R} = \frac{R/\sqrt{3}}{R};\ \tan\theta = \frac{1}{\sqrt{3}};\quad \theta = 30^o$$

$$Power\ factor(pf) = \cos(\theta);\ pf = \cos(30^o) = \frac{\sqrt{3}}{2}$$



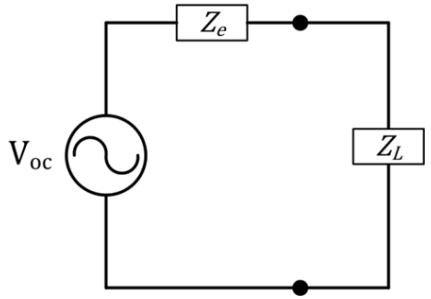
$$P = V_{rms} I_{rms}\cos(\theta) = \frac{(V_R)^2}{R} = \frac{(V_{rms}\cos(\theta))^2}{R}$$ (projecting V_S onto to the V_R axis – see phasor diagram above)

$$\therefore R = \frac{(120\cos(30^o))^2}{25} = 432\ \Omega;\quad \omega L = \frac{R}{\sqrt{3}};\ \therefore L = \frac{R}{\omega\sqrt{3}} = \frac{432}{2\pi\times 60\times\sqrt{3}} = 0.6616\ H$$

Maximum Power Transfer

- For resistive circuits, the power transfer is maximised when the load is made equal to the source resistance.
- We now wish to examine the power transfer for the case of an AC source driving a load impedance.

V_{oc} is the maximum amplitude



$$Z_L = R_L + jX_L$$

$$Z_e = R_{eq} + jX_{eq}$$

$$I_m = \frac{|V_{oc}|}{|Z_{eq} + Z_L|}$$

- We wish to find suitable conditions for the load parameter R_L and X_L that will maximise power (P_L) transfer to the load.

$$P_L = R_L I_{rms}^2; \quad P_L = R_L \left(\frac{1}{\sqrt{2}}\right)^2 \frac{|V_{oc}|^2}{(R_{eq}+R_L)^2+(X_{eq}+X_L)^2}$$

- On average, reactances absorb **zero** power and thus any non-zero values of $(X_{eq} + X_L)^2$ would only reduce P_L.
- We can easily eliminate this term by imposing $X_L = -X_{eq}$;

$$P_L = \frac{1}{2}|V_{oc}|^2 \frac{R_L}{(R_{eq} + R_L)^2}$$

- P_L is minimized when $\frac{dP_L}{dR_L} = 0 \Rightarrow R_L = R_{eq}$
- For maximum power transfer $R_L + jX_L = R_{eq} - jX_{eq}$; $Z_L = Z_{eq}^*$ Complex conjugate
- In words, power transfer from an AC source to a load is maximised, when the load impedance is made equal to the complex conjugate of the source impedance.
- The maximum average power $P_{L\,(max)}$

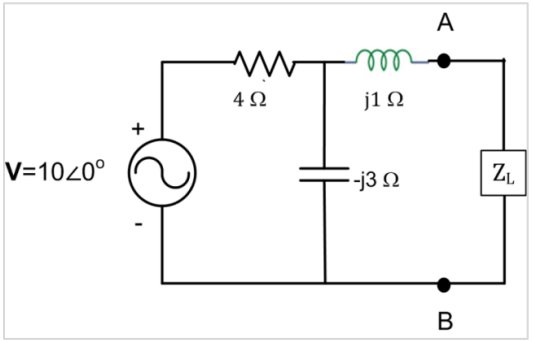
$$P_{L\,(max)} = \frac{1}{2}|V_{oc}|^2 \frac{R_{eq}}{(R_{eq} + R_{eq})^2}; \quad P_{L\,(max)} = \frac{1}{8}\frac{|V_{oc}|^2}{R_{eq}}$$

V_{oc} is the maximum amplitude

Examples

Example 4

❑ In the circuit shown below, find the load that will receive the maximum power.

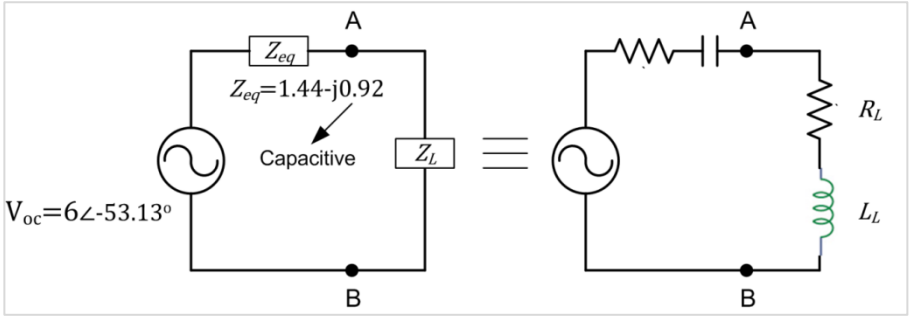


Solution

- Using Thevenin's theorem:

$$V_{AB} = \frac{-j3}{4-j3} \times 10\angle 0^o = 6\angle -53.13^o = V_{oc}$$

$$Z_{eq} = j1 + 4 \,||\, -j3 = j + \frac{4(-j3)}{4-j3}$$
$$= j1 - \frac{j12}{4-j3} = 1.44 - j.092\Omega$$



$$P_{L\,(max)} = \frac{1}{8}\frac{|V_{oc}|^2}{R_{eq}}$$

- For maximum power transfer

$$Z_L = Z_{eq}^* = 1.44 + j.092\Omega$$

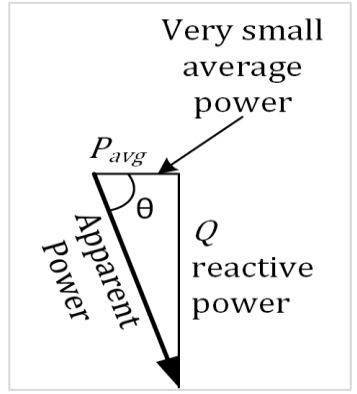
$$P_{L\,(max)} = \left(\frac{1}{8}\right)\frac{6^2}{1.44} = 3.125\ Watts$$

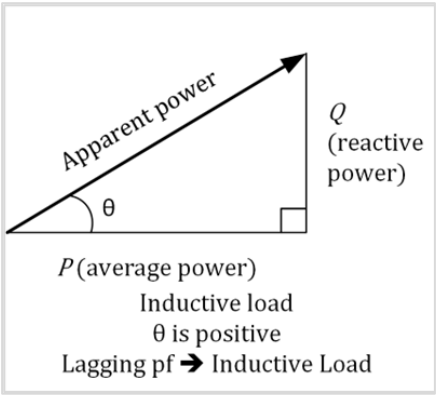
Power Factor Correction

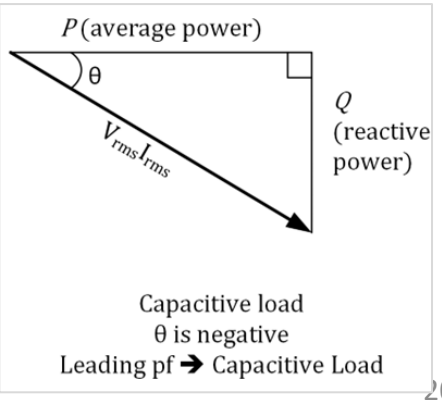
- Large currents can flow through energy storage devices (inductance and capacitance) without average power being delivered.
- In heavy industry, many loads are partially inductive and large amount of reactive power flow.

$$P_{average} = V_{rms} I_{rms} \cos\theta \quad Watts$$
$$Reactive\ Power\ (Q) = V_{rms} I_{rms} \sin\theta \quad VAR\ (reactive)$$
$$Apparent = V_{rms} I_{rms} \quad VA$$

- This reactive power causes higher currents in the power distribution system.
- Consequently, the lines and transformers must have higher ratings that would be necessary to deliver the same average power to a resistive (100% power factor) load.
- Energy rates are charged to industry depend on the power factor, with higher charges for energy delivered at low power factors.
- Remember power factor is not taken into account for residential customers.
- Therefore, it is advantageous to choose loads that operate at near unity power factor [θ= 0, cos(θ) = 1]
- A common approach is to place capacitors in parallel with inductive load to increase the power factor(i.e. decrease $\theta \Rightarrow \cos\theta$ increases, power factor increases).
- Note as $\theta \to 0$, $Q \to 0$ ➔ 100% power factor







Examples

Example 5

❑ A 50 kW load operates from a 60 Hz 10 kV rms line with a power factor of 60% lagging. Compute the capacitance that must be placed in parallel with the load to achieve a 90% lagging power factor.

Solution

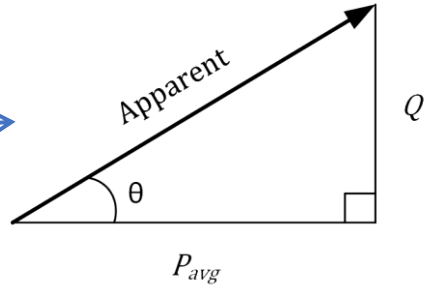
- Power angle calculation:

$\cos\theta = 0.6 \Rightarrow \therefore \theta = 53.13°$

$Q = P_{avg}(\tan\theta)$

$= 50 \times 10^3 \times \tan 53.13°$

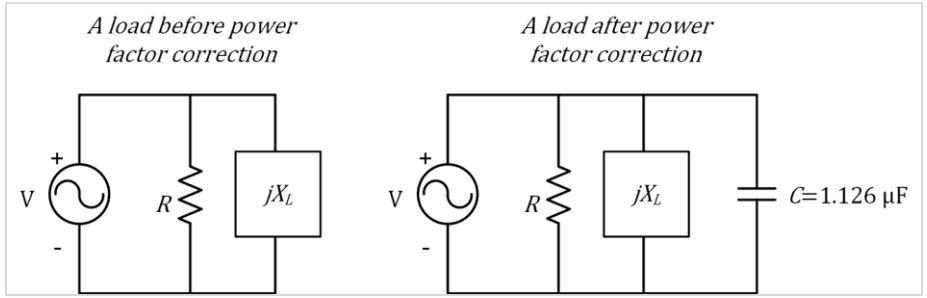
$= 66.67\ kVAR$



- After adding a capacitor, the power will still be 50 kW and he power angle become $\theta_{new} = cos^{-1}(0.9) = 25.84°$
- The new value of the reactive power will be: $Q_{new} = P_{avg} \times \tan(\theta_{new}) = 50 \times 10^3 \tan 25.84° = 24.22\ kVAR$
- ∴ The reactive power (Q_c) of the capacitance must be: $Q_C = Q_{new} - Q = 24.22 - 66.67 = -42.45\ kVAR$

$$Q_C = \frac{V_{rms}^2}{X_C} \Rightarrow X_C = \frac{V_{rms}^2}{Q_c} = \frac{(10\times10^3)^2}{42.45} 2356\ \Omega; \qquad \therefore C = \frac{1}{\omega X_C} = \frac{1}{2\pi\times60\times2356} = 1.126\ \mu F$$

Note: 60 Hz 10 kV rms line



Examples

Example 6

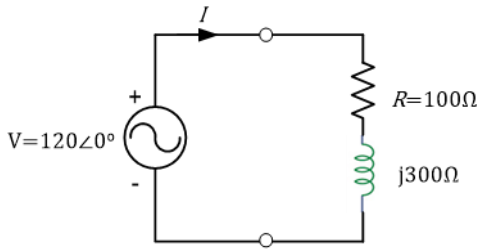
❑ (a) If a household AC voltage of $V_{rms} = 120\ V$ is applied to a load $z = 100 + j300\ \Omega$, find $P_{average}$, Q (reactive power), and apparent power. Assume $f = 60\ Hz$.

Solution

- $I_{rms} = \frac{V_{rms}}{\sqrt{R^2+X_L^2}} = \frac{120}{\sqrt{100^2+300^2}} = 0.3796\ A$
- $Apparent\ Power = V_{rms}I_{rms} = 120 \times 0.3795 = 45.54\ VA$

$$P_{average} = I_{rms}^2 \times R = (0.3795)^2 \times 100 = 14.40\ W$$

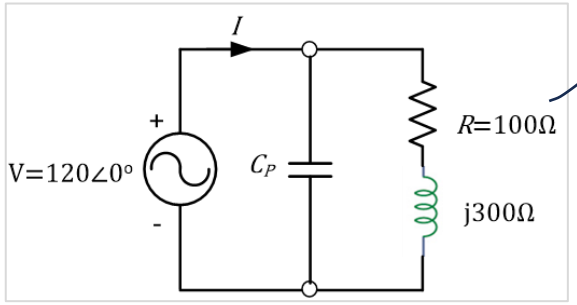
$$Q\ (Reactive\ Power) = I_{rms}^2 X_L = 300 \times 0.3795^2 = 43.20\ VAR$$

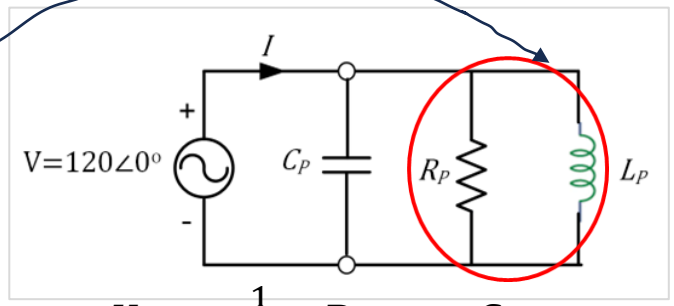


❑ Find a suitable reactive element that, when connected in parallel with the load in part (a) above will yield $pf = 1$ (i.e. $\theta = 0$, $\cos\theta = 1$).

$I = \frac{120\angle 0°}{100+j300} = 0.3795\angle -71.6°$; $\cos\theta = 0.316$ Lagging; To achieve $pf = 1$, we need a parallel capacitance C_p.

Series to parallel conversion





$$X_C = \frac{1}{\omega C_P};\ B_C = \omega C_P$$

Susceptance: $|B_L| = \frac{X_L}{R^2+X_L^2} = \frac{300}{100^2+300^2} = 0.003\ \Omega^{-1}$

For zero angle ($\cos\theta = 1$), the circuit must have only resistance, i.e. $B_C = |B_L|$; $\omega C_P = 0.003\Omega^{-1}$ $\therefore C_P = 0.003/2\pi \times 60 = 7.958\ \mu F$

ELEC2134: Circuits and Signals

AC Circuit Analysis

Tutorial questions

AC Power Analysis

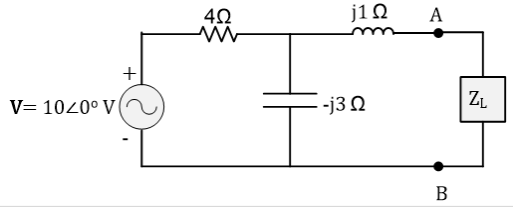
- ✓ AC Power
- ✓ Reactive Power
- ✓ Apparent Power
- ✓ Power Triangle
- ✓ Maximum Power Transfer
- ✓ Power Factor Correction

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Tutorial Questions (AC Power Analysis)

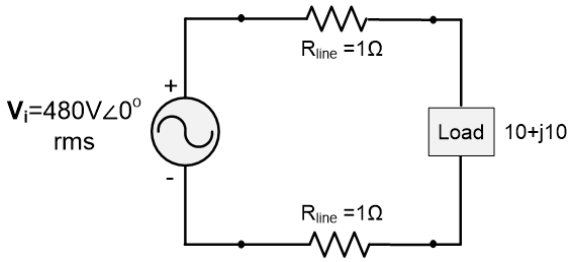
Question 1: For the circuit shown below, find the load that will receive the maximum power. What is the power?



Ans: $Z_L = 1.44 + j0.92\ \Omega$; $P_L\,(Max) = 17W$

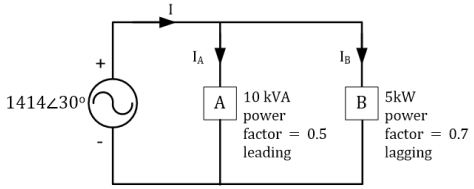
Question 2: For the circuit shown below, find:

a. The average power delivered to the load
b. The average power absorbed by the line
c. The apparent power supplied by the voltage source
d. The power factor of the load
e. The power factor of the line plus the load



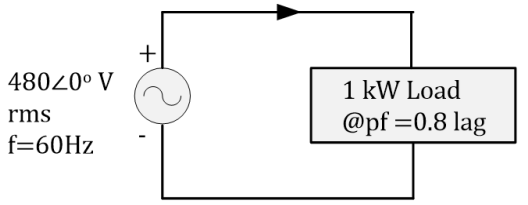
Ans: (a) 9.44 kW
(b) 1.89 kW
(c) 14.75 kVA
(d) 0.707 lag
(e) 0.768 lag

Question 3: A voltage source delivers power to two loads connected in parallel. Find the power, reactive power, and power factor for the source, also find the phase current I.



Ans: 10 kW, 3.559 kVAr, 0.942 lead, 7.507∠50° A

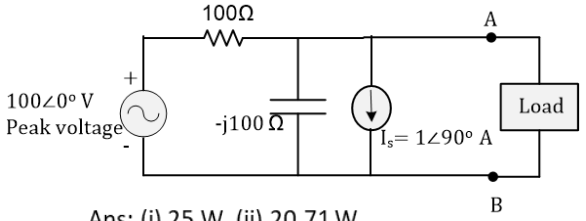
Question 4: It is desired to 'raise' the power factor of the load to 0.95 lagging for the circuit shown below. Compute the size and the rating of a capacitor which, when added across the load, will accomplish this.



Ans: 4.85 μF

Question 5: Determine the maximum power that can be delivered to a load for the circuit shown below if

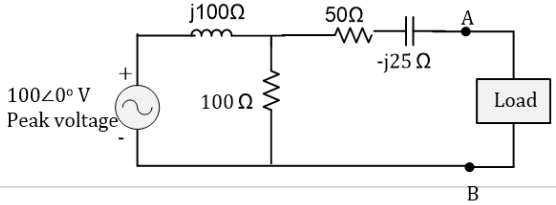
i. The load can have any complex value
ii. The load must be a pure resistance



Ans: (i) 25 W, (ii) 20.71 W

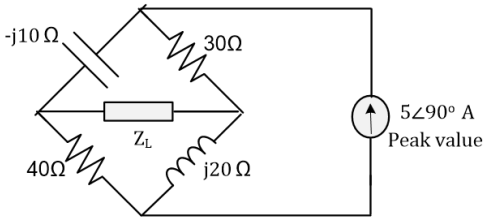
Question 6: Determine the maximum power that can be delivered to a load by the circuit shown below if

i. The load can have any complex value
ii. The load must be a pure resistance



Ans: (i) 6.25 W, (ii) 10.71 W

Question 7: Calculate the value of $\mathbf{Z}_L$ for the circuit given below in order for $\mathbf{Z}_L$ to receive maximum average power. What is the maximum average power received by $\mathbf{Z}_L$?



Ans: 20 Ω, 31.25 W